



Pakistan sign language recognition: leveraging deep learning models with limited dataset

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Abstract

Sign language is the predominant form of communication among a large group of society. The nature of sign languages is visual. This makes them very different from spoken languages. Unfortunately, very few able people can understand sign language making communication with the hearing-impaired extremely difficult. Research in the field of sign language recognition can help reduce the barrier between deaf and able people. A lot of work has been done on sign language recognition for numerous languages such as American sign language and Chinese sign language. Unfortunately, very little to no work has been done for Pakistan Sign Language. Any contribution in Pakistan Sign Language recognition is limited to static images instead of gestures. Furthermore, the dataset available for this language is very small in terms of the number of examples per word which makes it very difficult to train deep networks that require a considerable amount of training data. Data Augmentation techniques help the network generalize better by providing more variety in the training data. In this paper, a pipeline for the Pakistan Sign Language recognition system is proposed that incorporates an augmentation unit. To validate the effectiveness of the proposed pipeline, three deep learning models, C3D, I3D, and TSM are used. Results show that translation and rotation are the two best augmentation techniques for the Pakistan Sign Language dataset. The models trained using our data-augment-supported pipeline outperform other methods that only used the original data. The most suitable model is C3D which not only produced an accuracy of 93.33% but also has a low training time as compared to other models.

Keywords Sign language recognition · Pakistan Sign Language · PSL data dictionary · C3D · Data augmentation

1 Introduction

Sign language is a mode of communication used by people with hearing disabilities. Sign languages (SLs), in contrary to spoken languages, make use of the visual channel where the signs are produced using the body and perceived using the eyes. SLs are not international but are influenced by different regions and cultures. There are many sign languages that exist today, such as American Sign Language (ASL), British Sign Language (BSL), Chinese Sign Language (CSL), etc. According to the World Federation of the Deaf, there are over 300 sign languages used by 70 million deaf people around the world [1].

The nature of sign languages is visual, making them very different from spoken languages. Unfortunately, very few able people can understand sign language, making communication with the hearing-impaired extremely difficult. Research in sign language recognition (SLR) can help reduce the barrier between deaf and able people. Several SLR techniques have been developed. The earlier researchers performed SLR using direct measurements via data gloves and other sensors. Over the years, the trend has shifted to vision-based techniques due to the challenges faced in direct measurements. Several action recognition models have been employed to develop highly accurate SLR systems.

The basic building block of SL is a gloss, which is a transcription of a sign in spoken language [2]. In simple terms, a gloss is the label of the sign. Depending upon the number of glosses, the SLR can be categorized into two categories: single gloss or multiple glosses. The former is known as Isolated SLR (ISLR) while the latter is sentence-level or Continuous SLR (CSLR) [2]. Within ISLR, there is a Word-level SLR (WLSLR) in which the prediction made by the SLR system is

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that of a word. A sign in SL may be composed of hand shapes and movements, which are referred to as manual components [3]. In addition, a sign may also consist of facial expressions, head motion and body postures, which are referred to as non-manual components.

The native language of Pakistan's deaf community is Pakistan Sign Language (PSL). There are around 250,000 people in Pakistan with hearing disabilities [4]. Although much work has been done on the recognition of other sign languages like ASL, [5–7] and CSL [8–10], no significant contribution has been made for PSL Recognition (PSLR). Considering the number of Pakistanis with hearing disabilities, PSLR is extremely vital as it can be used by normal people to easily and effectively communicate with people with hearing disabilities. The only dataset available for PSL is the PSL Dictionary dataset [11]. Unfortunately, this dataset only has two samples per sign which is one of the main challenges in developing a reliable SLR system for PSL.

In this paper, a sign language recognition system for PSL is proposed that is based on the PSL Dictionary dataset. As sign language recognition is, in fact, an action recognition task, three action recognition models were trained on the PSL dataset: Convolutional 3D (C3D), Inflated 3D ConvNet (I3D), and Temporal Shift Module (TSM). As the deep learning models require a significant amount of data for training, which was not available in our case, several image augmentation techniques were implemented to artificially increase the data. The generation of synthetic data proved to be beneficial in the case of our small dataset. The basic motivation behind this research work is to achieve an accurate system for recognizing words signed using PSL.

1.1 Research contribution

The major contributions of this research are as follows:

1. First significant study on PSL recognition as an action recognition problem. The previous studies in PSLR are limited to static images.
2. First major contribution that uses the PSL dictionary dataset.
3. The deep learning models require a significant number of samples of each class for training. However, each sign in the PSL dataset has only two samples. So, an SLR pipeline is proposed that yields good accuracy on a limited dataset.
4. First study that employs and discusses the Temporal Shift Module (TSM) for sign language recognition.

The rest of the paper is organized as follows. Section 2 provides an overview of related work. The details of the PSL dataset are given in Sect. 3. Section 4 presents the proposed PSLR pipeline. Results are presented in Sect. 5 while the

discussion follows in Sect. 6. The conclusions and future directions are given in Sect. 7.

2 Literature review

Over recent years, various techniques have been developed for SLR. Formerly, SLR was done using direct measurements. Data were collected using different equipment like gloves [12, 13], sensors or other motion-capturing devices, for example, Kinect. However, amid different challenges associated with direct measurement techniques, the research trend has shifted to vision-based approaches. In vision-based approaches, the signs are captured using a camera and are stored in the form of images or videos. The image-based SLR approaches, for example, [14], are limited to only a few actions such as alphabet, numbers, and different daily use signs. On the other hand, the video-based SLR paves the way to train models on the entire sign language dictionary.

With the advancement of computer vision and deep learning, the utilization of Convolutional Neural Networks (CNNs) and deep neural networks for vision-based SLR has produced promising results. Earlier, 2D CNNs were used for SLR, but they could not incorporate temporal information. Later, 3D CNNs were employed with the capability of learning spatiotemporal features. One of the first works that introduced 3D CNN in SLR employed Convolutional 3D (C3D) [15] for ISLR [16]. Inflated 3D ConvNet (I3D) [17], mainly developed for video-based action recognition, has proved to be the state-of-the-art model in this regard as it has demonstrated outstanding performance in ISLR [2] and WLSLR [18]. A recent study in CSLR proposed a Multi-scale Local-Temporal Similarity Fusion Network (mLTSF-Net) for adaptively fusing local features via temporal similarity [19]. They convolved each scale of the selected frames via position-aware convolution for ensuring temporal consistency.

Researchers have also been interested in studying the performance versus computation cost trade-off between 2D CNNs and 3D CNNs. One way to save the computation cost is to decompose the 3D architecture into two architectures: 2D spatial convolution and 1D temporal convolution [20, 21]. A blend of 2D and 3D networks [22, 23] was studied in two ways: first using 2D and later 3D or first using 3D and later 2D. The two architectures are referred to as top-heavy and bottom-heavy, respectively. For either architecture, the high-level or low-level temporal modeling was removed. To overcome this computation cost, Lin et al. [24] presented a new perspective of efficient temporal modeling for better understanding and recognition of videos by proposing a novel Temporal Shift Module (TSM). The main idea was to achieve good accuracy at a low computational cost.

The Word-level SLR can be categorized into static and sequential. In static WLSLR, a word is represented by a fixed shape of hands, whereas in sequential WLSLR, a word is represented by the sequence of hand and arm movements. Two different WLSLR models based on holistic visual appearances and 2D human poses were proposed in [25]. They first employed transfer learning to extract spatial features from the given videos using VGG16 pre-trained on ImageNet. The extracted features were passed onto the gated recurrent unit (GRU). For the pose-based model, 50 body and hand points are selected using OpenPose and then fed to GRU. Apart from that, they also presented a novel pose-based temporal graph convolution network (Pose-TGCN), where the human body was considered a fully connected graph and a weighted adjacency matrix is created accordingly. The model was trained by passing this graph to the neural network.

Context is of key importance for action recognition. For SLR, the attention mechanisms like transformers [26] can better learn the features. Slimane et al. [27] worked on a word-based dataset to emphasize the context as well as the gesture while passing it into a training model. It was noted that although the dominant hand tends to carry significant information regarding the meaning of a sign, other features such as facial expressions and the surroundings also play a key role in identifying the correct meaning. According to the proposed pipeline, spatial features were extracted from the data frame through the 2D CNN model MobileNetV2 [28] which has low latency and high efficiency as a feature extractor. Then the model uses temporal modeling to understand temporal changes before learning the sequence alignment using the connectionist temporal classification [29]. Another study [30] used transformer network to classify the static signs where static images were used instead of sequences to represent words. They used OpenPose [31] to extract important landmarks such as joints and fingers. A total of 137 key points were extracted. 2D CNN, pre-trained on ImageNet, was also used separately to extract features. Relevant features from every frame were identified and fed to a transformer network consisting of a multi-head attention mechanism for predicting the static signs. Boháček and Hruš [32] analyzed the ability of the transformer model to recognize isolated sign language based on body-pose representation and proposed a computationally low-cost WLSLR system. The proposed pipeline begins with the extraction of landmarks using the vision API algorithm. They also augmented the data using in-plane rotation, squeeze, perspective transformation, and sequential joint rotation. The normalization process is performed to prevent the model from learning irrelevant spatial features. The proposed architecture is a modified transformer mentioned by Vaswani et al. [26]. The result showed a significant improvement compared to the previous state-of-the-art pose-based approaches. Kindiroglu et al. [33] proposed aligned temporal accumulative fea-

tures (ATAF) for the representation of isolated sign language videos using skeleton heatmap. Using accumulative representations, constant length feature can be used to represent videos with variable lengths. The extracted key frames were aligned using temporal transformer networks (TTNs), and then the descriptors were extracted using CNNs.

Apart from conv nets, several other techniques have been applied for SLR. Lim et al. [3] used hand tracking and hand representation using particle filter and hand energy image (HEI). Furthermore, Sincan and Keles [34] proposed an ISLR model based on Motion History Images (MHI) generated from RGB video frames, and without any segmentation of hands or face regions. Jiang et al. [35] used skeleton-based action recognition that mainly focuses on the distinctive patterns of human joint position and motion. Table 1 provides a summary of all SLR techniques discussed so far. They also employed a multi-modal approach capturing action data from different sources to further improve the performance.

Pakistan Sign Language (PSL) is the native language of Pakistan's deaf community. Although, a PSL recognition system is essential for this community and its dataset (i.e., dictionary) is publicly available [11], no significant work has been done to cater to this need. The research work done so far is either based on static images or supported by a limited dataset consisting of alphabet or digits. One of the earlier researchers used statistical template matching to recognize PSL gestures [38]. The primary input device in their case was DataGlove5. Raees et al. [39] proposed an image-based PSLR system where deep pixels-based analysis and Template Matching were used to for recognizing fingers and thumb, respectively. Afterward, the isolated signs were recognized by the position and direction of the fingers whether they are lowered or raised. A gesture-based PSLR system was proposed in [40] that uses a leap motion device to acquire the training data. The data are then fed into a correlation algorithm for sign recognition.

Malik et al. [41] used a webcam to capture the input data, then segmented the hand from the background, and extracted the required features using Principal Component Analysis (PCA). Finally, the signs were classified using K Nearest Neighbors (KNN). Naseem et al. [42] proposed a framework for PSL translation into speech and text using CNN. Although [43] acquired input data from videos, their SLR technique was image based. They extracted four vision-based features, that is, local binary patterns, a histogram of oriented gradients, an edge-oriented histogram, and speeded-up robust features. These features were then classified using Multiple kernel learning (MKL) in SVM. Javaid et al. [44] proposed a video-based action recognition system for PSL using Inflated 3D ConvNet (I3D) and Transformer network. An SLR system for PSL static alphabet used Faster R-CNN for hand localization and symmetric mean-based binary patterns (sMBP) for feature extraction [45]. The signs were finally classified

Table 1 A summary of SLR techniques

Technique	Dataset	Best acc	References
Convolutional 3D (C3D)	UCF-101	52.8%	[15]
Inflated 3D ConvNet (I3D)	HMDB-51, UCF-101	80.9%, 98.0%	[17]
I3D with transfer learning	MS-ASL, SIGNUM	Up by 21%	[36]
VGG16, GRU, Pose-TGCN	Word-Level American Sign Language	62.63%	[25]
2DCRNN, 3DCNN	MoSL	92%, 99%	[37]
SSD, 2DCNN, SVD, LSTM	RKS-PERSIANSIGN, first-person, ASVID, isoGD	99.5%, 91%, 93%, 86.1%	[15]
CNN hand tracker (CNNT) with hand energy image (HEI)	RWTH-BOSTON-50	89.33%	[3]
Motion History Images (MHI) with 3DCNN	AUTSL, BosphorusSign22k	93.53%, 94.83%	[34]
Skeleton aware multi-modal SLR framework (SAM-SLR)	AUTSL (RGB and RGB-D track)	98.42%, 98.53%	[35]
Sign Attention Network (SAN)	RWTH-PHOENIX-Weather 2014	29.7%	[15]
Transformer networks with OpenPose	Flemish Sign Language	74.7%	[30]
Transformer networks with vision API	WLASL, LSA64	63.18%, 100%	[32]

Table 2 A review of PSLR techniques

Technique	Dataset	Video based action recognition	References
Statistical template matching	Alphabet	No	[38]
Deep pixels-based analysis and template matching	Alphabet and digits	No	[39]
Leap motion device	No information	No	[40]
Principal Component Analysis (PCA) and K Nearest Neighbors (KNN)	Alphabet	No	[41]
Convolution neural network	Digits	No	[42]
Multiple kernel learning (MKL), SVMs	Alphabet	No	[43]
Inflated 3D ConvNet (I3D) and transformer	Basic expressions	Yes	[44]
Faster R-CNN, symmetric mean-based binary patterns (sMBP), and error correction output codes (ECOC)-based SVMs	Alphabet	No	[45]
Bag-of-Words (BoW) and SVMs	Alphabet	No	[46]

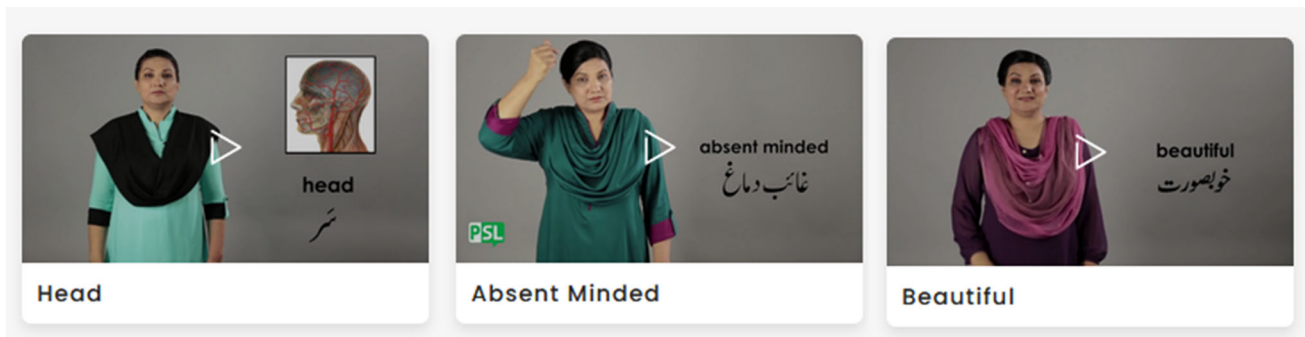


Fig. 1 Snapshot of three samples from the PSL dataset

using error correction output codes (ECOC)-based SVMs. Another image-based PSLR system was proposed that used Bag-of-Words (BoW) and SVMs [46]. The input data were preprocessed and segmented using Thresholding. Table 2 provides a summary of all PSLR techniques. The features were extracted using Speeded Up Robust Feature (SURF) and the obtained SURF descriptors were clustered using K-means clustering.

3 Dataset

For Pakistan SLR, the PSL dictionary dataset [11] was used. This dataset contains signs for over 6000 words. However, only 80 signed words were taken from this dataset that was listed in the top 1500 most frequently used words in Urdu. This list was taken from the list of 5000 most frequent words compiled by the Centre For Language Engineering [47]. Most of the signs (63) are performed by a single female signer with a different set of clothing. The word being signed is also exhibited in Urdu and English as shown in Fig. 1. In some cases, a supporting image is also provided.

The rest of the 17 signs are performed by two male and one female signer. The dataset is in the form of videos. Each word has exactly one video in which a signer performs a particular sign twice. In each video, a signer performs a gesture and brings the arms to the default hanging position before performing the same gesture again.

The signed words fall under different categories such as verbs, adjectives and body parts. The distribution of words across different categories is given in Table 3.

4 Methodology

This section presents the proposed pipeline for the recognition of PSL that incorporates a data augmentation unit. The main concern with the PSL dataset is that there are only two samples per word which makes it very difficult to develop

Table 3 Count of words of each type in our dataset

Type	Total samples
Adjectives	15
Adverbs	12
Body anatomy	5
Family and marriage	7
Numbers	5
Prepositions	7
Pronouns	13
Transport	6
Verbs	10
Total	80

a robust and reliable SLR system. The augmentation unit aims to alleviate this shortcoming by synthetically generating more data. Before the data can be augmented, it must be cleaned and then split into training and testing data since only the training data need to be augmented. The detailed methodology is shown in Fig. 2. Different deep learning models are employed to determine and evaluate the impact of the augmentation techniques.

4.1 Preprocessing

As mentioned in Sect. 3, every word has exactly one video in which a signer performs a particular sign exactly two times. We, therefore, had to split each video into two samples. Furthermore, words being signed are captioned in both English and Urdu text. In some cases, an image is also provided (see Fig. 1). In the preprocessing stage, the text and images were also removed. This was done automatically by cropping the videos since the text or images have no overlap on the vertical axis. Figure 3 shows the resulting video frame for the word “heart” after preprocessing.

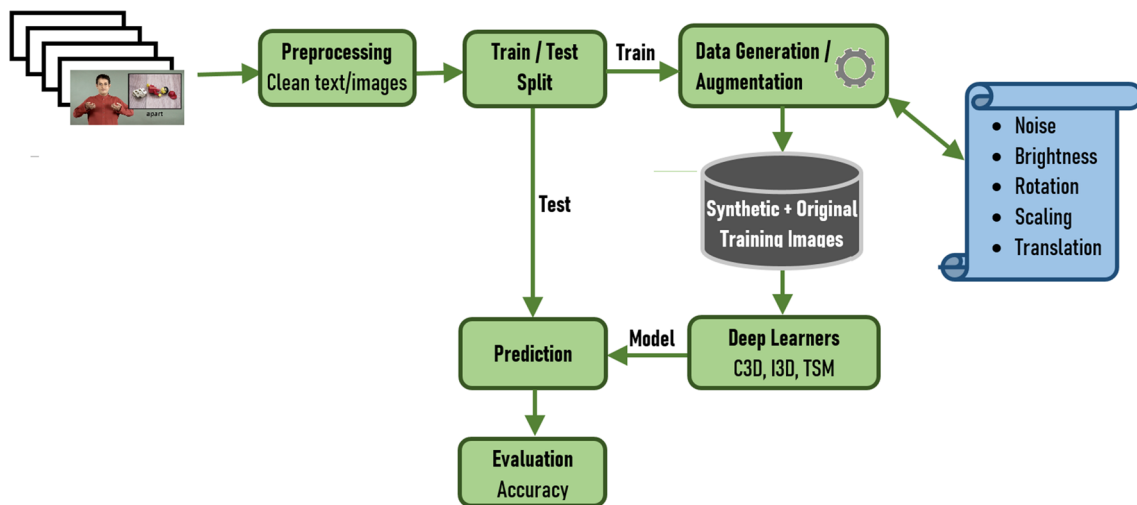


Fig. 2 Proposed pipeline for PSLR

Fig. 3 Before and after cropping: the frame is taken from the sign “heart”

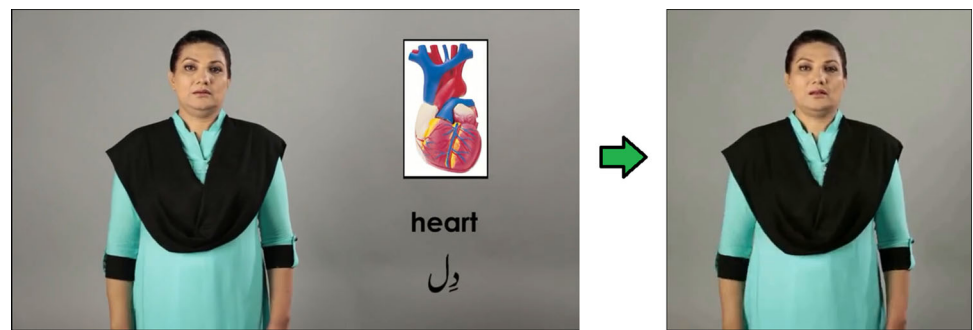


Table 4 Number of samples

Type	Total samples	Training	Testing
PSL dataset	160	80	80

4.2 Training testing split

Once the data have been cleaned, it must be split into training and testing data. Since every class or sign has just two samples, one sample is used for training and the other sample is set aside for testing. Table 4 highlights the number of training and testing samples used for experiments.

4.3 Data augmentation

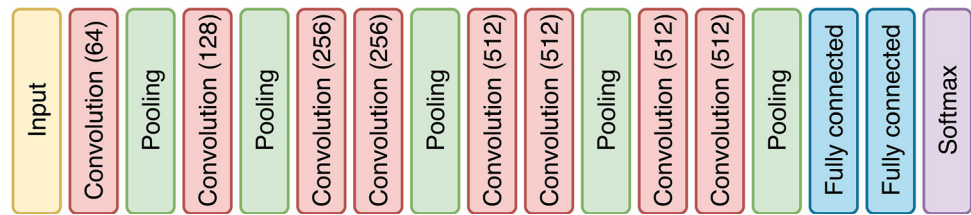
Data augmentation techniques are commonly used by researchers [48] since collecting new data is some times not possible, or too expensive. Image augmentation approaches can be used to tackle over-fitting as they can help the training network generalize better. Since each sample is in the form of a video, it was augmented by augmenting all of its frames one at a time.

In this paper, five geometric and color-manipulation augmentation techniques were employed to increase the size of the training examples. These were brightness (and contrast), noise, rotation, scaling, and translation. These techniques were implemented using the OpenCV-Python library with the following parameters: rotation_angle = 10, scale_percent = 75, and translation_factor = 10. The brightness was controlled by control variables alpha and beta with values of 1.5 and 5, respectively. The salt and pepper noise was added with a proportion of 0.05.

It is worth mentioning here that data augmentation was only performed on the training data and no changes were made to the test data. The test data were used to evaluate the effect of augmentation on the sign prediction task.

4.4 Deep learning models

Both machine learning (ML) and deep learning (DL) techniques have been utilized for isolated SLR. However, as discussed in Sect. 2, most recent studies have only used DL models that have been shown to be more effective. More specifically, CNNs have been utilized for developing SLR systems. Therefore, two CNN architectures namely, Convolutional 3D (C3D) and Inflated 3D ConvNet (I3D) were used

Fig. 4 Various layers in C3D**Table 5** Input and output dimensions of each layer of C3D as shown in Fig. 4

Layers	Input data size	Output data size
C	$480 \times 480 \times 3 \times 80$	$64 \times 478 \times 478 \times 1 \times 78$
P	$64 \times 478 \times 478 \times 1 \times 78$	$64 \times 239 \times 239 \times 1 \times 39$
C	$64 \times 239 \times 239 \times 1 \times 39$	$128 \times 237 \times 237 \times 1 \times 37$
P	$128 \times 237 \times 237 \times 1 \times 37$	$128 \times 118 \times 118 \times 1 \times 18$
C	$128 \times 118 \times 118 \times 1 \times 18$	$256 \times 59 \times 59 \times 1 \times 9$
C	$256 \times 59 \times 59 \times 1 \times 9$	$256 \times 57 \times 57 \times 1 \times 7$
P	$256 \times 57 \times 57 \times 1 \times 7$	$256 \times 28 \times 28 \times 1 \times 3$
C	$256 \times 28 \times 28 \times 1 \times 3$	$512 \times 26 \times 26 \times 1 \times 1$
C	$512 \times 26 \times 26 \times 1 \times 1$	$512 \times 24 \times 24 \times 1 \times 1$
P	$512 \times 24 \times 24 \times 1 \times 1$	$512 \times 12 \times 12 \times 1 \times 1$
C	$512 \times 12 \times 12 \times 1 \times 1$	$512 \times 10 \times 10 \times 1 \times 1$
C	$512 \times 10 \times 10 \times 1 \times 1$	$512 \times 8 \times 8 \times 1 \times 1$
P	$512 \times 8 \times 8 \times 1 \times 1$	$512 \times 4 \times 4 \times 1 \times 1$
F1	$8192 = (512 \times 4 \times 4 \times 1 \times 1)$	4096
F2	4096	4096
Softmax	4096	80

C: Convolutional, P: pooling and F: fully connected layers

for developing the PSL recognition system. In addition, Temporal Shift Module (TSM) was also employed for SLR for the first to test its ability to be deployed for an SLR system.

4.4.1 Convolutional 3D (C3D)

Convolutional 3D (C3D) [15] are deep 3-dimensional convolutional networks (3D ConvNets) trained on a large-scale supervised video dataset such as Sports-1M dataset [49]. The model was proposed to learn the spatio-temporal features. These learned features, when combined with a simple linear classifier, yielded good results on different video analysis tasks. Although 3D ConvNets were proposed before [49, 50], the C3D model was built using modern deep architectures and was employed to large-scale supervised training datasets that resulted in the best performance on various video analysis tasks.

The C3D network contains 8 convolutional layers, 5 pooling layers, 2 fully connected layers, and a softmax output layer as shown in Fig. 4. The number of 3D convolution filters in each convolutional layer is represented in the respective box. Each filter is $3 \times 3 \times 3$ with stride $1 \times 1 \times 1$ for both spatial and temporal dimensions. All 3D pooling filters are $2 \times 2 \times 2$ with stride $2 \times 2 \times 2$. Both fully connected layers have 4096

output units. Table 5 shows the input and output dimensions of each layer of C3D. The input videos were normalized to $480 \times 480 \times 3 \times 80$.

4.4.2 Inflated 3D ConvNet (I3D)

Inflated 3D ConvNet (I3D) [17] is considered state-of-the-art model for action recognition. It was developed to improve C3D model by inflating all the filters and pooling kernels of the 2D architecture. The filters in the 2D model are square $N \times N$. After inflation, they become cubic $N \times N \times N$. By doing so, they were able to add time as the third dimension which facilitated the learning of the spatio-temporal information. As compared to the previous network architectures like LSTM, 3D-ConvNet, and Two-Stream, they proposed a Two-Stream 3D-ConvNet architecture as shown in Fig. 5.

The 2D filters from pre-trained models were used to bootstrap 3D filters. In other words, the researchers used the parameters from a 2D model trained on a very large dataset (for example, ImageNet) in their 3D architecture. This bootstrapping was done by repeating an image N times. By doing so, they repeated the 2D filters N times along the time dimension, and then rescaled them by dividing the N . Although a 3D ConvNet can learn motion features from RGB inputs directly,

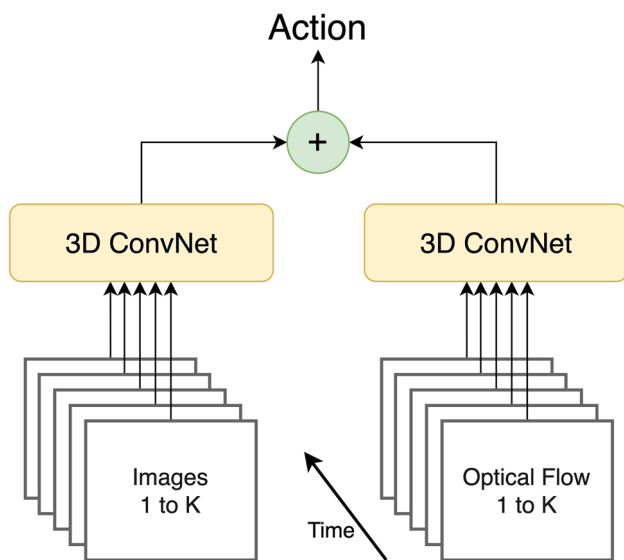


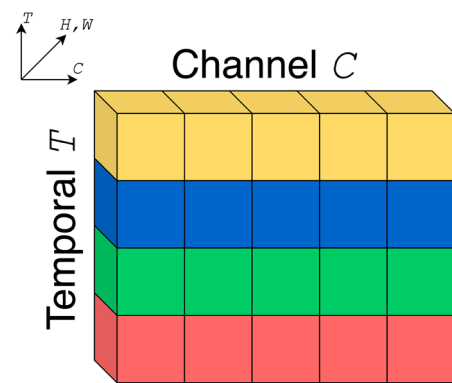
Fig. 5 Two-stream 3D-ConvNet

it only performs feedforward computation. On the other hand, an optical flow architecture is somewhat recurrent. Therefore, the authors suggested a Two-Stream network, where one 3D network was trained on RGB data and the other on the optical flow inputs. These two networks are trained separately and their predictions are averaged at the test time.

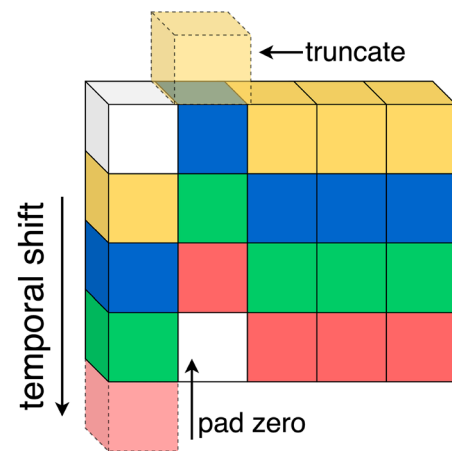
4.4.3 Temporal shift module (TSM)

Lin et al. [24] presented a new perspective of efficient temporal modeling for better understanding and recognition of videos by proposing a novel Temporal Shift Module (TSM). The main idea was to achieve a good accuracy at a low computational cost. The traditional 2D CNNs are computationally efficient but do not model the temporal information. On the other hand, 3D CNNs produce good results, but at the expense of high computational cost which makes its deployment expensive on the edge devices. Contrary, the TSM can produce results similar to those of 3D CNNs while maintaining the complexity of 2D CNNs. Therefore, the TSM is somewhat a hybrid model that achieves a good accuracy with high efficiency.

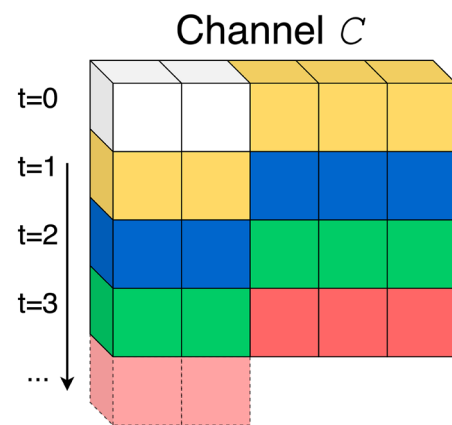
The 2D CNNs require no temporal modeling (Fig. 6a) as they operate independently over the dimension T . On the other hand, the proposed TSM moves the channels along the temporal dimension in both directions. Two TSM networks were proposed: uni-directional TSM and bi-directional TSM. The bi-directional TSM (Fig. 6b) was proposed for offline video recognition that achieved state-of-the-art performance. The main idea was to combine the past and future frames with the current frame. The bi-directional TSM operates as follows: Initially, a set of T frames are sampled from the input video. Subsequently, each frame is processed individually



(a) The original tensor without shift



(b) Offline temporal shift (bi-direction)



(c) Online temporal shift (uni-direction)

Fig. 6 The high-level diagram explaining the TSM networks

by 2D CNNs. Finally, the resultant outputs are aggregated through averaging to give the final prediction. For real-time online video understanding, uni-directional TSM (Fig. 6c) was proposed that only shifted the past frame because future frame cannot be shifted to the present.

5 Results

5.1 Experimental setup

We used MMAction2 [51] to train and test the C3D, I3D, and TSM models. The training was done for 50 epochs with a learning rate of 0.001 for C3D and 0.01 for I3D and TSM. All models used ReLU activation function and a dropout ratio of 0.5. The training and testing were performed using Google Colab. Each experiment (training and testing) was run three times. Due to the limited amount of training data, we did not have any validation set. We also did not tune the hyperparameters and used the default values specified in [51]. The accuracies reported in this paper are testing accuracies and are the average of the accuracies of the three runs.

5.2 Evaluation metrics

The most significant metric for this study is accuracy. Since the dataset is very evenly balanced, accuracy is the appropriate measure. Accuracy is the percentage of correctly predicted samples from all samples in the dataset as given in Eq. 1. Metrics such as recall and precision are most suited for medical data but also used here.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Number of Samples}} \quad (1)$$

A prediction is considered correct if the top prediction, i.e., the suggestion with the highest probability matches the word in the actual/test example.

Usually, for a two-class dataset where one class can be considered as positive and the other as negative, recall (or sensitivity) and precision are given by Eqs. 2 and 3.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3)$$

where TP is True Positive, FP is False Positive, FN is False Negative and TN is True Negative.

However, for a multi-class dataset such as the one used in this study, precision and recall are calculated for every class c . The precision and recall for class c are given by Eqs. 5 and 4.

$$\text{Precision}_c = \frac{\text{TP or correct prediction of class } c}{\text{All predictions labeled as class } c} \quad (4)$$

$$\text{Recall}_c = \frac{\text{TP or correct prediction of class } c}{\text{Number of actual class } c \text{ samples}} \quad (5)$$

Table 6 The effect on the size of training data when augmentation techniques are used separately, and when training data are augmented using all five techniques

Augmentation	Training (org. + aug.)	Testing
Brightness (B)	160 (80 + 80)	80
Noise (N)	160 (80 + 80)	80
Rotation (R)	160 (80 + 80)	80
Scaling (S)	160 (80 + 80)	80
Translation (T)	160 (80 + 80)	80

Table 7 Comparison of C3D, I3D, and TSM using original training data

Model	Top acc.	Top-3 acc.	Top-5 acc.
C3D	66.67	89.58	97.08
I3D	77.50	94.58	98.33
TSM	33.75	60.83	75.83

5.3 Data augmentation results

In this work, five augmentation techniques: brightness, noise, rotation, scaling, and translation were employed. Figure 7 shows some samples of augmented images.

5.4 Effect of geometric and other augmentation techniques on data size

We applied five augmentation techniques to the original training data only. First, each augmentation technique was applied separately to the training data producing the same number of samples as the training data. Then different augmented samples were combined together along with the original training data. The total number of training samples after each augmentation method is given in the second column in Table 6 where org. and aug. represent original and augmented, respectively. The test set was not augmented.

5.5 Effect of augmentation on PSL recognition

All three deep learning models, C3D, I3D, and TSM, were run on the original and augmented datasets. Beginning with the original dataset, Table 7 compares the performance of the three models on original, non-augmented training data. Column 2 shows the accuracy when the actual word matches the top predicted word, while columns 3 and 4 indicate the top-3 and top-5 accuracies, respectively. In top- n accuracy, the actual word needs to be in the top n predicted words to count as the correct prediction. In this study, all experiments were run three times, so each value represents a mean accuracy. The best values are shown in bold. When only the original data is used, I3D has the best accuracy.

Fig. 7 Samples of augmented images using each of the 5 techniques

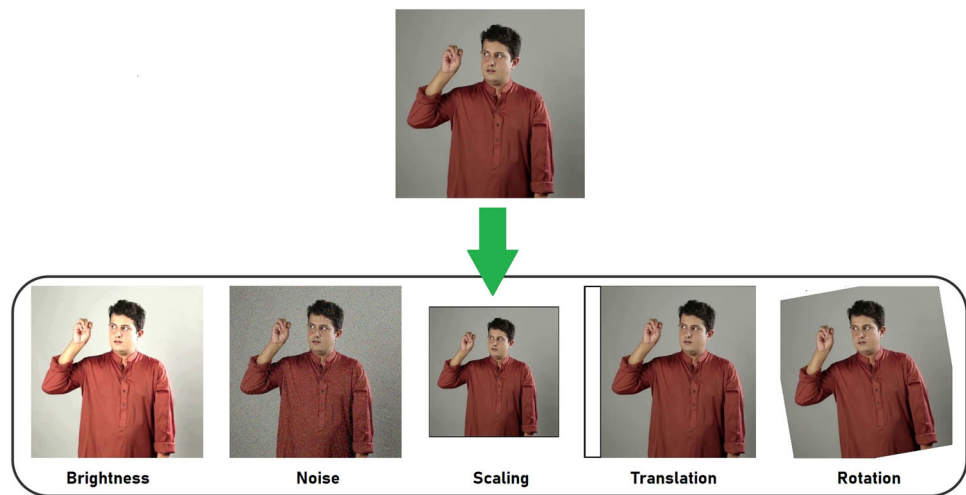


Table 8 compares the effect of the five augmentation techniques on the performance of the three models. The original training data consisted of 80 samples, however, when data augmentation was used, training data consisted of 160 shuffled real and synthetic examples. During the evaluation, each of the 80 test examples was fed to each of the three trained models. A prediction is considered correct if the top prediction, i.e., the suggestion with the highest probability matches the word in the actual/test example. Table 8 also shows the top-3 and top-5 accuracies. The reason for including these accuracies in the results is mainly because top- n accuracy is commonly used in word prediction applications in natural language processing. The best values are shown as bold whereas underlined values represent the second best accuracies.

From Table 8, it can be seen that T has the best effect on C3D and I3D while S produces the best results for TSM. The second-best accuracies are shown as underlined. Generally, T and R seem to be good candidates for data augmentation for PSLR. This can also be confirmed from the top-5 accuracy results where both of these methods become more prominent. Comparing with Table 7, C3D and I3D show improved performance when data augmentation is utilized. Interestingly, TSM's performance decreases when the training data are increased. Overall, C3D has the best results for PSLR.

5.5.1 Combining data of two augmentation techniques

So far, one augmentation technique has been used at a time to increase the training data. What would be the impact of combining the data of two augmentation techniques? For five techniques, there are ten different combinations to examine. However, using Table 8 as a reference, these combinations can be narrowed down. Note that the two best augmentation techniques for C3D are B and T, for I3D these are R and T, whereas for TSM these are R and S. These three combina-

Table 8 Comparison of five augmentation techniques in terms of their impact on the performance of the three models

Augmentation	C3D	I3D	TSM
<i>Top-1 accuracy</i>			
Brightness (B)	<u>87.50</u>	67.50	24.17
Noise (N)	75.00	70.00	24.17
Rotation (R)	85.83	79.17	<u>25.83</u>
Scaling (S)	82.92	69.17	27.08
Translation (T)	90.42	79.17	25.42
<i>Top-3 accuracy</i>			
Brightness (B)	99.17	86.67	48.75
Noise (N)	97.50	90.00	45.42
Rotation (R)	<u>98.75</u>	95.00	<u>52.50</u>
Scaling (S)	97.08	88.75	52.92
Translation (T)	97.92	<u>94.17</u>	50.42
<i>Top-5 accuracy</i>			
Brightness (B)	100.00	94.17	67.50
Noise (N)	99.17	96.67	67.50
Rotation (R)	99.17	97.50	72.50
Scaling (S)	99.58	94.58	68.33
Translation (T)	<u>99.58</u>	<u>96.67</u>	<u>69.17</u>

In each case, the training data consists of 80 original + 80 augmented samples

tions can be used. Furthermore, since T produced the best results for C3D and I3D, and S produced the best results for TSM, the combination of S and T was also considered. Using two techniques would increase the training data to 240 samples. Table 9 shows the impact of such an increase on the performance of the models.

Recall that the best accuracies for C3D, I3D and TSM were 90.42%, 79.17% and 27.08%, respectively, (refer to Table 8) when only one augmentation technique was used. From Table 9, it can be seen that the accuracies for all three

Table 9 The effect of combining the data of two augmentation techniques with the original training data

Augmentation	C3D	I3D	TSM
B + T	92.08	78.33	21.67
R + S	91.25	82.08	32.50
R + T	93.33	87.50	35.83
S + T	92.92	83.75	21.88

models improved when the data from two augmentation techniques were used with R and T methods, producing the best results for all models. Scaling combined with T yielded the second-best accuracies for the C3D and I3D. Since each experiment was done three times, each value indicates mean accuracy. In addition, the standard deviation for C3D and I3D was low indicating that the results are statistically significant.

Table 10 shows the precision and recall of the proposed SLR system using C3D for each class when R and T augmentation techniques were used along with the original data.

Figures 8, 9 and 10 present the confusion matrices of C3D, I3D, and TSM for the original and R + T augmented training data. Since there is only one test example per class, so each combination of predicted and actual values has a value of 1 (shown as green) or 0 (shown as red). It can be concluded that data augmentation using R and T has positively affected the performance of the deep learning models and rotation and translation are the two best augmentation approaches for PSLR.

5.5.2 Further experiments with R and T augmentation techniques

We have so far concluded that R and T are the best techniques to augment PSL data. In this section, limiting the discussion to these two techniques, a few more experiments were performed. These experiments were conducted to investigate whether or not the accuracy can be improved even further. Since TSM did not perform well, it was excluded from these experiments.

In the first experiment, the signer is segmented from the background. Initially, the videos had a plain background with varying shades of gray. To segment the signer, the body segmentation library BodyPix [52] was used which is an open-source library that performs person and body-part segmentation. The segmented signers were placed on a uniform black background. In the second experiment, all alternate frames from all videos were eliminated to reduce the size of the data and speed up the training process. Finally, in the third experiment, both R and T were applied in each video of the original training data. As a result, 80 augmented samples were obtained, instead of 160 if R and T had been applied separately. The results of these experiments are shown in

Table 10 Precision and recall for every class using C3D with R and T augmentation

Class	Precision	Recall	Class	Precision	Recall
100	1	1	Mother	1	1
10	1	1	Mouth	1	1
50	1	1	Near	1	1
Able	1	1	Off	0	0
Absolutely	1	1	Often	1	1
According	1	1	one	1	1
All	1	1	Outdoors	1	1
Almost	1	1	Outside	0.5	1
Ancient	1	1	Parents	1	1
Annual	1	1	Request	1	1
Another	1	1	Say	1	1
Any	1	1	She	1	1
Baby	1	1	Sister	1	1
Because	1	1	So-accentuator	1	1
Both	1	1	So-in-order-to	1	1
Brain	1	1	Some	0	0
Bus	1	1	Soon	1	1
Car	1	1	Street	1	1
Come	1	1	Sudden	1	1
Continuously	1	1	Sufficient	1	1
Cycle	1	1	There	1	1
Do	1	1	This	1	1
Dry	0	0	Thorough	1	1
Empty	1	1	Tongue	1	1
Eye	1	1	Travel	1	1
Father	1	1	Truck	1	1
Few	0.5	1	True	1	1
From	1	1	Universal	1	1
Go	1	1	Up	1	1
He	1	1	Urgent	1	1
Heart	1	1	Very	1	1
Help	1	1	Walk	1	1
However	1	1	Warm	1	1
I	1	1	We	1	1
Keep	0.5	1	Weak	0.5	1
Lakh	1	1	Whole	1	1
Literally	1	1	without	1	1
Make	0	0	Woman	1	1
Many	1	1	Write	1	1
Marriage	1	1	You	1	1

Table 11. For comparison, the base accuracies obtained for R and T techniques (as shown in Table 9) are also given.

From Table 11, it can be seen that modifications such as segmenting signer from background and using alternate

Fig. 8 Confusion matrices of C3D when trained on **a** Original data and **b** R + T augmented data

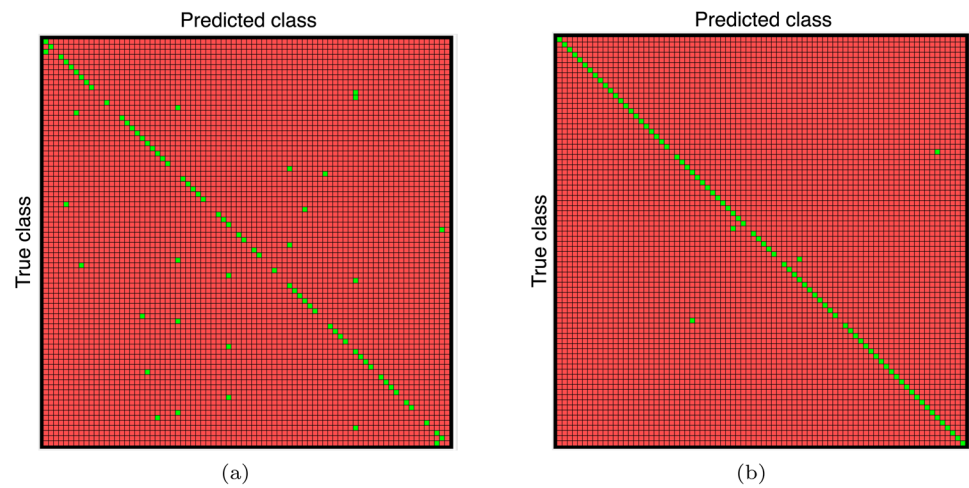


Fig. 9 Confusion matrices of I3D when trained on **a** Original data and **b** R + T augmented data

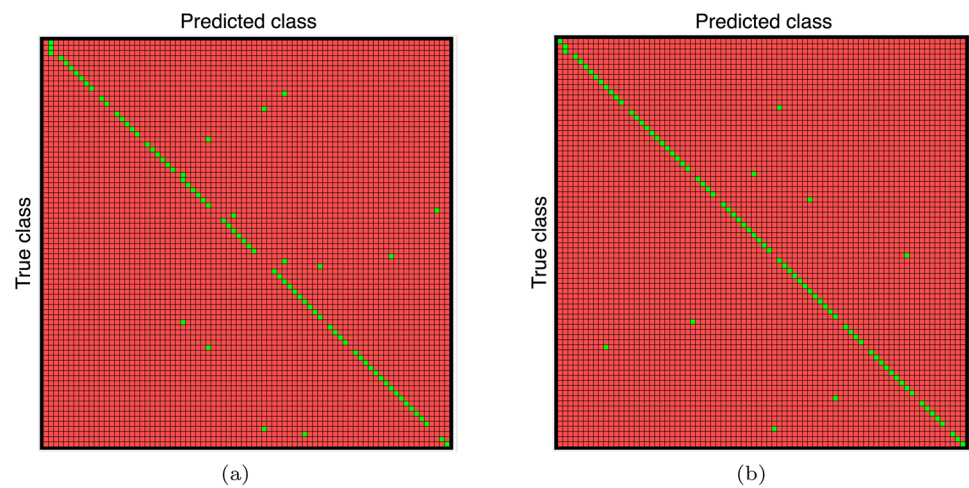


Fig. 10 Confusion matrices of TSM when trained on **a** Original data and **b** R + T augmented data

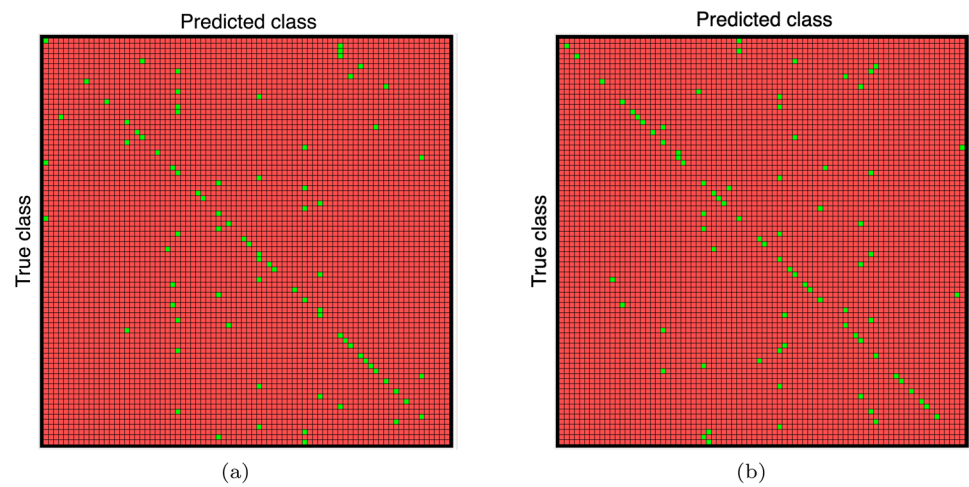


Table 11 Results of further experiments using R and T augmentation techniques only

Experiment	Samples	C3D	I3D
Base accuracies (R + T)	240	93.33	87.50
Signer segmentation	240	84.17	80.83
Alternate frames	240	88.75	21.67
RT on each video	160	86.25	77.92

frames. adversely affected the performance of the deep learning models.

5.6 Response of the system on signs not included in the dataset

In the preceding sections, experiments were conducted on testing data that exclusively comprised signs present in the training dataset. This particular section is devoted to examining the response of the proposed system when confronted with signed words that were not included in the training dataset. To accomplish this, the confidence values generated by our model were leveraged to establish a threshold for prediction outcomes. Through experimentation and observation, it was determined that a threshold of 6.5 on the confidence values of the C3D model could be set. Subsequently, any prediction value falling below this threshold value was categorized as “unknown.”

An experiment was conducted on the C3D model trained using R + T augmentation. To assess its performance on the signs not included in our dataset, ten random signs from the PSL dictionary that were not present in the training dataset were selected. The results of this experiment are displayed in Table 12 under the “Signs not included in the training dataset” heading. For the purpose of comparison and comprehension, data of the signs from the training dataset were also included, specifically the ten signs with the lowest confidence values.

Table 12 Comparison of how well the proposed PSLR system recognizes the original signs versus signs not included in the dataset

Signs included in training dataset				Signs not included in the training dataset			
Actual	Predicted	Confidence	Result	Actual	Predicted	Confidence	Result
Off	Outside	7.94	Outside	Scissors	Come	7.36	Come
We	We	7.80	We	Monsoon	Outside	6.81	Outside
Any	Any	7.68	Any	Asteroid	Sufficient	6.12	Unknown
Few	Few	7.32	Few	Politician	Truck	6.08	Unknown
Because	Because	7.26	Because	Satellite	Come	5.76	Unknown
This	This	7.10	This	Curtains	Come	5.53	Unknown
Without	Without	7.06	Without	Intern	Woman	5.45	Unknown
Do	Do	6.94	Do	Winter	Many	5.06	Unknown
Some	Few	6.83	Few	Tea	Absolutely	3.40	Unknown
From	From	6.76	From	Grasshopper	Bus	2.90	Unknown

The confidence represents the confidence value of the C3D model

This information is presented in the table under the “Signs included in the training dataset” heading. Upon inspection, it was observed that the confidence value for all the signs included in the training data was above the established threshold. Consequently, it can be concluded that none of the 80 signed words included in the training dataset would be classified as ‘unknown’ by our system. On the other hand, when analyzing the confidence values of the signs not included in the training dataset samples, in most of the cases (8 out of 10), the established threshold value of 6.5 was able to predict unknown signed words as ‘unknown’.

6 Discussion

In this paper, a pipeline for word-level SLR is proposed for Pakistan Sign Language that has a very limited number of samples per class. In fact, there are just two examples per class. As part of the pipeline, an augmentation module is incorporated for generating more training data. For this purpose, a mix of five geometric, noise and color-manipulation augmentation techniques are used. This research uses three deep learning models: C3D, I3D and TSM.

When the 80 original training samples were used for training, C3D and I3D produced the best accuracies of 66.67% and 77.50%, respectively. TSM only had an accuracy of 33.75%. When the three models were trained using data from one augmentation technique, the performance of the first two models improved as shown in Table 8. However, the accuracy of TSM dropped to as low as 24.17%. TSM has been used successfully for action recognition but underperformed for PSLR. The reason for this is that TSM relies on optical flow that captures motion in a sequence of frames. In the PSL dataset, the initial arm and hand movement is the same for all words. Similarly, after the sign is performed, the movement of the arm to its sideways resting position is also similar for all words. The movement of the hand that is actu-

ally performing the sign is the only distinguishing feature for every word but it is concentrated across a few frames. As a result, the reliance on movement makes it difficult for TSM to learn the significant patterns of signed words. When the data are augmented even further so that the training data is increased to 240 samples, TSM registered an improvement but that too was incomparable to the performance of C3D (93.33%) and I3D (87.50%) as shown in Table 9. With two augmentation techniques, all three models produced the best results for the R + T combination. In conclusion, it is R and T are suggested as the two best augmentation approaches for PSLR. All experiments in this study were done three times, so all accuracies, etc. indicate mean values. In addition, the standard deviation, especially for C3D and I3D, was low indicating that the results are statistically significant.

With R and T as the best techniques to augment PSL data, a few more experiments on R and T augmented datasets were conducted: (1) segmenting signer (2) using alternate frames (3) applying both R and T on each video. From Table 11, it can be seen that all of these additional experiments adversely affected the performance of the models. Some artifacts were introduced when the body parts were detected and segmented using BodyPix. These artifacts could be one of the reasons for the decline in models' performances. Eliminating alternate frames certainly decreased the training time, but it also reduced the accuracy. As mentioned earlier, the critical information that distinguished signed words from one another was concentrated in mostly the center frames. Therefore, further elimination of frames resulted in the loss of important information.

In this study, the three deep learning models were also trained by combining the data from five augmentation techniques with the original one, producing a dataset of 400 samples. The results of this experiment are given in Table 13. Furthermore, Fig. 11 compares the accuracies and training times for original training data, R + T augmented training data and B + N + R + S + T augmented training data. The training time for C3D is relatively less than that of the other two models, possibly because of its simpler architecture and fewer parameters. I3D typically requires more computational resources than C3D due to its inflated architecture, which involves pre-training 2D models and inflating the weights to 3D. The growth of data size with the use of more augmentation techniques is depicted in Fig. 12. These figures indicate that using all five augmentation techniques instead of R + T significantly increased the training time and data size.

The performance of C3D improved, but that improvement is not significant considering the accuracy vs. training time and data size trade-off. With five augmentation techniques, the accuracy of I3D was reduced which could be because some augmentation techniques affected the training of the model. From Table 8, it can be observed that these could be B, N, or S as their accuracy was lower than that of the original

Table 13 The effect of combining the data of all augmentation techniques with the original data

Augmentation	Samples	C3D	I3D	TSM
R + T	240	93.33	87.50	35.83
B + N + R + S + T	400	94.58	85.00	57.08

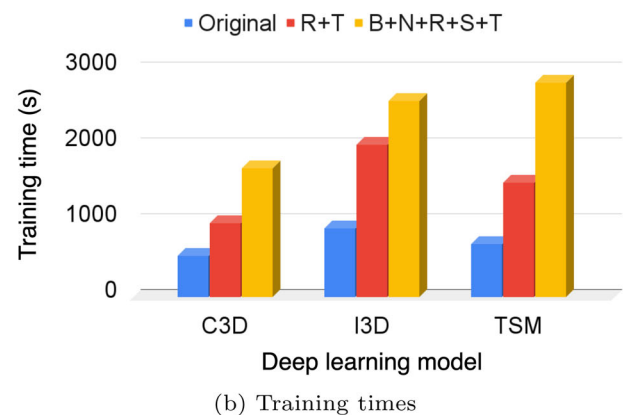
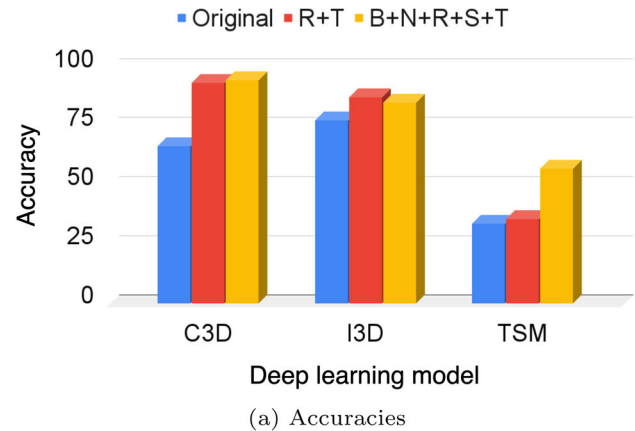


Fig. 11 Impact of increasing the augmented data on accuracy and training time

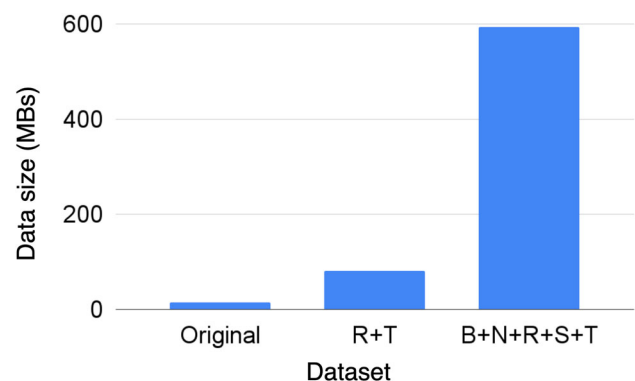


Fig. 12 Growth of data size by employing more augmentation techniques

data. The improvement in the results of TSM was significant which shows that the performance of TSM improves with the size of the training data.

7 Conclusion and future work

This paper presents the first major study on PSL recognition. Previous studies only focused on static images instead of videos and hand/arm gestures. Also, this is the first study that utilizes the PSL Dataset in developing an SLR system. In this dataset, a gesture or sign of a word is performed only twice, so there are not a lot of training examples per word to train deep learning models. Data Augmentation can help and in this paper, the benefits of augmentation techniques were exploited. A pipeline for PSLR was proposed with an in-built data augmentation module. The pipeline was incorporated with three deep learning models and within the augmentation module, five techniques were used. The best augmentation techniques for PSLR are rotation and translation and when the original training data was augmented using the data from these two techniques, all models produced the best accuracy. Overall, the best model is C3D. TSM that has been used successfully for action recognition failed for PSLR. When data from all five augmentation techniques were used to train the models, the mean accuracy improved for C3D, but based on a statistical test, these results were not significant.

Some of the future directions for this works are as follows:

1. Extend the number of classes.
2. Try different data augmentation techniques.
3. Online/real-time system.

Finally, in order to truly eliminate the communication barrier between the able and hearing-impaired, a text-to-sign or speech-to-sign translation system can be developed for PSL such as [53, 54] that were developed for Indian sign language.

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Declarations

Ethical standards This statement is to certify that the author list is correct. The Authors also confirm that this research has not been published previously and that it is not under consideration for publication elsewhere. On behalf of all Co-Authors, the Corresponding Author shall bear full responsibility for the submission. There is no conflict of interest. This research did not involve any human participants and/or animals.

Conflict of interest The authors declare that they have no relevant financial or non-financial interests to disclose. There is no personal relationship that could influence the work reported in this paper. No funding was received for conducting this study. The authors have no conflicts of interest to declare that are relevant to the content of this article.

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